# Dr. Pradipta K. Banerjee

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Dept. of CSE-AIML
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## Education

2015 **Post Doctoral Fellow, Tsinghua University, Beijing**

*Title*: Facial Landmark Tracking for Analysis of Human Neuro-psychological Factors
*Supervisor* : Prof. Yu-Jin Zhang, Department of Electronic Engineering

Tenure : 2 years (not completed)

2014 **PhD, University of Calcutta**

*Title*: Some Studies on Performance Improvement of Frequency Domain Correlation Technique for Face Recognition
*Advisor*: Prof. Asit K. Datta

2002-2004 **Master of Technology (Electrical Engineering), University of Calcutta**

*Title*: Multi-resolution Analysis in Block Pulse Function Domain using C
*Advisor*: Prof. A. Deb, Prof. G. Sarkar
*79.5%, 1st class*

1999-2002 **Bachelor of Technology (Electrical Engineering), University of Calcutta**
*Title*: PC Based Transfer Function Identification using Block Pulse Function
*Advisor*: Prof. A. Deb, Prof. G. Sarkar
*75.2%, 1st class*

1996-1999 **Bachelor of Science (Physics Hons.), Vidyasagar College, University of Calcutta**, *62.38%, 1st class*

## Publications

### Book

- Asit K. Datta, Madhura Datta, **Pradipta K. Banerjee**, *Face Detection and Recognition: Theory and Practices*, CRC Press, Taylor and Francis, 2015.

## Book Chapters

1. Papia Banerjee, **Pradipta K. Banerjee**, Asit K. Datta, *Efficient Image Plane Rotation Invariant Frequency Domain Face Recognition Technique Using Eye Localization*, Springer-Verlag Berlin Heidelberg, 2013
2. **Pradipta K. Banerjee**, Asit K. Datta, *Techniques of Frequency Domain Correlation for Face Recognition and its Photonic Implementation*, Ed.: A. Quaglia and C. M. Epifano, pp. 165-186, Face Recognition: Methods, Applications and Technology, *Series: Computer Science, Technology and Applications, NOVA Publishers*, 2012
3. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Class Dependent 2D Correlation Filter for Illumination Tolerant Face Recognition*, M.K. Kundu et al. (Eds.): *Lecture Notes in Computer Science, Springer-Verlag Berlin Heidelberg, 7143, 338-345*, 2012

## SCI/Scopus Journals

1. **Pradipta K. Banerjee**, Tamal K. Mukherjee, *Hybrid regression and explainable AI for phase transition analysis of two-flavour quark matter*, *Journal of Subatomic Particles and Cosmology, Elsevier, Article 100247*, 2025, https://doi.org/10.1016/j.jspc.2025.100247
2. S. Panja, A. Halder, S. Bhattacharya, **Pradipta K. Banerjee**, P. Mahato et al., *Deep Facial Feature Fusion and Voting Strategies for Enhanced Emotion Recognition*, *Procedia Computer Science, Elsevier, 260, 415-423*, 2025 **(Scopus)**
3. **Pradipta K. Banerjee**, Asit K. Datta, *Bandpass correlation filter for illumination and noise tolerant face recognition*, *Signal Image Video Processing, Springer, 11, 1, 9-16*, January 2017, **(SCI)**
4. **Pradipta K. Banerjee**, Asit K. Datta, *Class specific subspace dependent nonlinear correlation filtering for illumination tolerant face recognition*, *Pattern Recognition Letters, Elsevier, 36, 177-185*, 15 January 2014, **(SCI)**
5. **Pradipta K. Banerjee**, Asit K. Datta, *A preferential digital-optical correlator optimized by Particle Swarm technique for multi class face recognition*, *JOLT, Elsevier, 50, 33-42*, 2013, **(SCI)**
6. **Pradipta K. Banerjee**, Asit K. Datta, *Generalized Regression Neural Network Trained Preprocessing of Frequency Domain Correlation Filter for Improved Face Recognition and its Optical Implementation*, *JOLT, Elsevier, 45, 217-227*, 2013, **(SCI)**
7. **Pradipta K. Banerjee**, Asit K. Datta, *An Illumination Tolerant Class Specific 2D Subspace Based Face Recognition Technique Using Optimum Correlation Filter*, *Optik - International Journal for Light and Electron Optics, Elsevier, 124, Issue 17, 3173-3179*, September 2013, **(SCI)**
8. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *A frequency domain face recognition technique based on correlation plane features as input to a regression neural network*, *Procedia Computer Science, Elsevier, 2, 75-82*, 2010, **(Scopus)**
9. Jayanta K. Chandra, **Pradipta K. Banerjee**, Asit K. Datta, *Neural network trained morphological processing for the detection of defects in woven fabric*, *Journal of Textile Institute, Taylor Francis, 101, 699-706*, 2010, **(SCI)**
10. **Pradipta K. Banerjee**, Asit K. Datta, *Illumination and noise tolerant face recognition based on eigen-phase correlation filter modified by Mexican hat wavelet*, *Journal of Optics, Springer, 38, 160-168*, 2008, **(SCI)**

## International / National Conferences

1. D. Chakraborty, **Pradipta K. Banerjee**, S. Datta, *Impact of Learning Rate Schedulers on ResNet Optimization for Facial Emotion Recognition*, International Conference on Communication and Computational Technologies (ICCCT), Springer, pp. 51-64, 2025
2. **Pradipta K. Banerjee**, *Phase-Encoded Cascaded Autoencoders based Correlation Filters and Adaptive Thresholding for Illumination Invariant Face Recognition*. In: Antonacopoulos, A. et al. (eds) Pattern Recognition. ICPR 2024. Lecture Notes in Computer Science, vol 15324, pp. 210-226. Springer, Cham. https://doi.org/10.1007/978-3-031-78383-8_12, 2025
3. D. Chakraborty, **Pradipta K. Banerjee**, *Facial emotion recognition using unconstrained minimum average correlation energy filter*, AIP Conference Proceedings, 2915(1), 2024
4. D. Chakraborty, U. Sur, **Pradipta K. Banerjee**, *Random forest based fault classification technique for active power system networks*, IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE), 2019
5. T. Ghosh, **Pradipta K. Banerjee**, *Linear Regression Correlation Filter: An Application to Face Recognition*, Computer Vision and Image Processing, Springer, IIITDM Jabalpur, 2019
6. T. Ghosh, **Pradipta K. Banerjee**, *A Class Specific Representation Learning for Illumination Tolerant Face Recognition*, Recent Trends in Image Processing and Pattern Recognition, IAPR (endorsed), Solapur University, 2018
7. A. Achuthanunni, R. K. Saha, **Pradipta K. Banerjee**, *Unconstrained Band-pass Optimized Correlation Filter (UBoCF): An application to face recognition*, IEEE Applied Signal Processing Conference (ASPCON), Jadavpur University, December 7-9, 2018
8. Papia Banerjee, **Pradipta K. Banerjee**, Asit K. Datta, *Face Detection and Landmark Localization using Non-parametric Approach and Landmark Filters*, *The Eighth International Conference on Advances in Pattern Recognition, IEEE Computational Intelligence Society*, Jan 4-7, 2015
9. Papia Banerjee, **Pradipta K. Banerjee**, *Phase-Eigen Subspace based Illumination Invariant Face Recognition using Associative Memory*, *IEEE National Conference on Computing and Communication Systems*, 2012, India
10. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Feature Based OT Parameter Selection of Correlation filter for Real Time Face Authentication in Frequency Domain*, *ICCCS, ACM, NIT Rourkela*, Odisha, India, 2011
11. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Face Detection and Eye Localization in Video by 3D Unconstrained Filter and Neural Network*, A. Abraham et al. (Eds.): *ACC 2011, Part II, CCIS 191, pp. 480-489, Springer-Verlag Berlin Heidelberg*, 2011, India
12. **Pradipta K. Banerjee**, Asit K. Datta, *An Unconstrained Frequency Domain Correlation Filter for Face Recognition with Reduced Misclassification Rate*, *Proceedings of International Conference on Trends in Optics and Photonics*, Dec 7-9, 2011, Kolkata, India
13. **Pradipta K. Banerjee**, *An approach of eye detection under varying in-plane rotation of face image*, *International Conference on Recent Trends in Information Systems, IEEE Computer Society*, 2011, India
14. Jayanta K. Chandra, **Pradipta K. Banerjee**, Asit K. Datta, *Singular value decomposition method for the detection of defect in woven fabric refined by morphological operation*, *IEEE Recent Advances in Intelligent Computational Systems*, 2011, India
15. Jayanta K. Chandra, **Pradipta K. Banerjee**, Asit K. Datta, *Modular eigen fabric based classifier for the detection of defect in woven fabric*, *IEEE Recent Trends in Information Systems*, 2011, India
16. Jayanta K. Chandra, **Pradipta K. Banerjee**, Asit K. Datta, *PCA and ANN based morpho-*

*logical process for the detection and classification in woven fabric*, *International Conference on Soft-computing and Engineering Applications*, 2011, India

17. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Wavelet Modified Phase-only Maximum Average Correlation Height Filter for Face Recognition*, *Proceedings of the Second International Conference on Intelligent Human Computer Interaction, Springer*, 2009, IIIT Allahabad, India
18. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *SVM Classifier for Face Recognition Based on Unconstrained Correlation Filter*, *IEEE International Conference on Signal and Image Processing Applications*, 2009, Malaysia
19. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Illumination Invariant Face Recognition: Comparison of Performances of Fourier Domain Correlation Filters*, *Proceedings of International Conference on Trends in Optics and Photonics*, 2009, Kolkata, India
20. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *A Study of Frequency Domain Correlation Filter in Face Recognition*, *National Conference on Modern Trends in Electrical Engineering*, 2009, Hooghly, India
21. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Optical Implementation of Face Recognition System Based on Unconstrained Minimum Average Correlation Energy Filter*, *Proceedings of International Conference on Trends in Optics and Photonics*, December 13-17, 2008, IIT Delhi, India
22. Jayanta K. Chandra, **Pradipta K. Banerjee**, Asit K. Datta, *Morphological Reconstruction Operation for the Detection of Defects in Woven Fabric*, *IEEE Region 10 Colloquium and the Third ICIIS*, 2008

# Research Interest

1. Broadly: Machine Learning, Deep Neural Networks, Image Processing and Pattern Recognition
2. Specifically: Multimodal Emotion Analysis and ATR

# Salient Features in Research Experience

- Regression correlation filters
- Class specific representation learning
- Correlation filter (classifier) development, performance evaluation and modification towards pattern recognition and object detection
- Performance improvement of correlation filters with different image processing applications under different illumination conditions, expression variations and noisy conditions of facial images
- Nonlinear correlation filtering in class specific subspace approach

# Research skills

1. Working platform: Python, Matlab, LaTeX
2. Algorithms developed: Machine Learning Algorithms, Deep Neural Networks, Correlation Filters, Regression Classifiers in Frequency Domain, Subspace Methods, Representation Learning, Clustering Algorithms

# Professional Experience

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| 2005–till date | **Future Institute of Technology, Kolkata / Future Institute of Engineering and Management, Kolkata**<br>Professor & Head, Department of CSE-AIML (Jan 2022 – till date)<br>Professor & Head, Department of Electrical Engineering (July 2005 – Dec 2021)<br>*Courses taken (Theory)*: Machine Learning, Deep Learning, Pattern Recognition, Computer Vision<br>*Courses taken (Practical)*: Machine Learning, Deep Learning |

# Teaching Experience

21+ years

# Professional Development Activities

## *Funded Project*

- **Position**: Principal Investigator
  **Title of Project**: Automatic Traffic Detection and Counting on Large Scale Real-Time Database on various road types in Kolkata and Howrah Municipal Corporation
  **Organisation**: WBPCB and The Energy and Resources Institute, Delhi
  **Year**: 2023-24
  **Amount**: Rs. 4.5 Lacs
  **Status**: Completed

## *Institution's Innovation Council (IIC), MoE Innovation Cell*

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| 2025–till date | **President**, Institution's Innovation Council, Future Institute of Technology, Kolkata |
| 2026 | **Organizer** : Smart India Hackathon (SIH) 2026, institute-level edition |
| 2025 | **Organizer** : DRISHTI 1.0 |
| 2025 | **Organizer** : Poster Competition and Innovation Challenge 2025 |

## *Resource Person / Invited Talks*

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| 10th – 15th Nov 2025 | **Resource Person**, AICTE Training and Learning (ATAL) Academy Faculty Development Programme on "Med-Tech Challenges and Opportunities under the 'Make in India' Framework", Future Institute of Engineering and Management, Kolkata (Certificate No. ATAL/2026/1781173287) |
| 11th July 2025 | **Invited Workshop**, "Machine Learning in Physics", Physics Pulse 2025, Department of Physics, Adamas University, Kolkata |
| 27th – 28th July 2023 | **Invited Speaker**, Physics Pulse, two-day symposium of the Department of Physics, Adamas University, Kolkata |

## *Project Developed (National Recognition)*

- **Position**: Mentor / Team Guide
  **Title of Project**: A small scale quadcopter proto for real time activity detection and classification
  **Responsibilities**: Developed state-of-the-art detection and classification algorithm using deep neural network in Python
  **Organisation**: Defence Research Unmanned System Exposition (DRUSE), organized by DRDO
  **Year**: 2018
  **Status**: Completed (Finalist)
- **Position**: Mentor / Team Guide
  **Title of Project**: A small-scale quadcopter proto for real time activity detection, classification and localization using deepnet and hybrid correlation filtering
  **Responsibilities**: Developed state-of-the-art detection, classification and localization algorithm
  **Organisation**: Aero India Student Pavilion, Bangalore
  **Year**: 2019
  **Status**: Bagged $6^{th}$ position
- **Position**: Mentor / Team Guide
  **Title of Project**: Automatic Smart Dustbin: A neural network and image processing application, ***Swachh Bharat Abhiyan Project***
  **Responsibilities**: Developed Convolutional Neural Network (CNN) to classify biodegradable, biomedical and nonbiodegradable objects using image processing techniques in Python
  **Organisation**: INAE Youth Conclave, organized by IIT Kharagpur
  **Year**: 2018
  **Status**: Bagged First Position

## *Ph.D Supervision*

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| Position: | Principal Supervisor |
| Ph.D Scholar: | Debosmita Chakraborty |
| Date of Enrollment: | 05.10.2020 |
| Title: | Some Studies on Multimodal Deep Learning Techniques and Frequency Domain Analysis for Enhanced Human Emotion Recognition and Mental Health Assessment |
| University: | MAKAUT, West Bengal |
| Progress: | Scopus indexed Journal (1), Conferences (2) |

## *Laboratory Development (Interdisciplinary)*

1. **Developed IoT (Internet of Things) project laboratory, based on Espressif and Raspberry Pi systems**
   (a) Smart home automation system through mobile app development
   (b) Controlling devices through Firebase and MQTT protocol
   (c) Both local and remote device control over WiFi
2. **Developed Machine Learning and Computer Vision Laboratory**

(a) Both CPU and GPU applications through Google Cloud
(b) Object tracking system with activity classification

## *Membership*

- Association for Computing Machinery (**ACM**)
- Computer Society of India **(CSI)**
- International Association of Computer Science and Information Technology (**IACSIT**)
- International Association of Engineers (**IAENG**)

## *Reviewer*

- International Journal of Pattern Recognition and Artificial Intelligence (IJPRAI)
- IEEE Signal Processing Letters
- Engineering Applications of Artificial Intelligence, Elsevier Press
- Computer Vision and Image Understanding, Elsevier Press
- International Conference on Science, Engineering and Technology
- IEEE Conference on Recent Trends in Information Systems

### *Workshops / Seminars / Conferences Organized*

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| 6th January 2026 | **Organizer** : Workshop on Computer Vision, Future Institute of Technology, Kolkata |
| 15th September 2025 | **Organizer** : Workshop on Statistical Machine Learning, Future Institute of Technology, Kolkata |
| 24th – 25th March 2019 | **Convener** : Two-day seminar on "Computational Intelligence in Signal Power and Control" (CISPC'19), F.I.E.M. Kolkata |
| 22nd – 23rd March 2018 | **Convener** : Two-day seminar on "Computational Intelligence in Signal Power and Control" (CISPC'18), F.I.E.M. Kolkata |
| 23rd – 24th March 2017 | **Convener** : Two-day seminar on "Computational Intelligence in Engineering Applications" (CIEA'17), F.I.E.M. Kolkata |

### *Training Programmes Attended*

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|---|---|
| 24th Jun – 5th Jul 2013 | Training Programme on "Digital Signal Processing, Speech & Image Processing", organized by C-DAC Kolkata as part of the Advanced Faculty Training Programme, HRD Division, Govt. of India (two weeks) |
| 30th Jan – 10th Feb 2012 | AICTE sponsored Staff Development Programme on "Optimization of Decision Tools using Soft Computing Models & MATLAB", Kalyani Govt. Engineering College |

### *Certifications*

- **DeepLearning.AI (Coursera)** : Introduction to TensorFlow for Artificial Intelligence, Machine Learning and Deep Learning (97.73%, 2022); Convolutional Neural Networks in TensorFlow (98.12%, 2022); Supervised Machine Learning: Regression and Classification (99.6%, 2022); Neural Networks and Deep Learning (96%, 2018); Improving Deep Neural Networks:

Hyperparameter Tuning, Regularization and Optimization (95%, 2018)

- **Stanford University (Coursera)** : Machine Learning (93.7%, 2016)

## Honours and Awards

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| 1998–1999 | Bela Rani Medal, Sudhir Kumar Bhose Memorial Medal, Niladri Bhattacharyay Memorial Award, H. L. Mitter Memorial Prize and Manada Sundari Stipend — for academic proficiency in B.Sc. (Physics Hons.), Vidyasagar College |

## Summary

Currently working as Professor & Head in the Dept. of CSE-AIML and President of the Institution's Innovation Council at F.I.T., Kolkata, with 21 years of teaching experience. My research interests include machine learning, neural networks and pattern analysis. I have published one (1) book, three (3) book chapters, seven (7) SCI journals, three (3) other peer-reviewed journal papers and twenty-two (22) international and national conference papers. For quick reference about my academic performance and research works, please visit my web page https://sites.google.com/view/pradiptakbanerjee

*I hereby declare that the details and information given above are complete and true to the best of my knowledge*

Signature